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Mapping the Cube to a Graph and Encoding



Graph Encoding



Figure 4. Encoding of all cube states onto 21 qubits.

Classical Divide-and-Conquer Approach

Figure 5. (a)–(c): Possible solutions for one pair of rows; (d): Final solution obtained by combining (a) and (c).

Figure 6. Encoding of all states onto 12 qubits, with each edge as a qubit per cube.

QUBO and Hamiltonian from Graph Constraints

- $$C_{graph} = (2x_1 + x_2 + y_1 + y_2 + z_1 + w_1 - 2)^2 + (x_3 + y_1 + z_1 + z_2 + w_2 - 2)^2 \\ + (x_2 + y_3 + 2z_3 + w_1 + w_3 - 2)^2 + (x_3 + y_2 + y_3 + z_2 + w_2 + w_3 - 2)^2$$

We can set the QUBO as

$$QUBO = C_{\text{cube}} + C_{\text{graph}}.$$

Dicke State Preparation and Associated XY-Mixers



As we prepared the tensor product of Dicke states as the initial state, we implement a special mixer that preserves the Dicke subspace $\{|001\rangle, |010\rangle, |100\rangle\}$. In this way, the constraints are preserved: $|001\rangle$, $|010\rangle$, and $|100\rangle$ rotate (mix) by β through the mixer.

First Iteration and Post-Processing



- **Approach 1:** Add the bias term of the first solution with zero cost $\{w_3, z_2, y_1, x_2\}$. In this approach, we apply a penalty strength of 5 and set the QAOA circuit depth to $p = 15$.

$$C_{bias} = 5(x_2 + y_1 + z_2 + w_3)^2, C_{graph \text{ (2nd iteration)}} = C_{graph \text{ (1st iteration)}} + C_{bias}$$

- **Approach 2:** Delete the solution variables $\{w_3, z_2, y_1, x_2\}$ from the previous C_{graph} , since edges should not be reused. Another solution will then be selected from the remaining edges.

Results



For the final check, with the first solution: Fig.5(a) and the second solution: Fig.5(c), we verified that no edges are shared. Therefore, the correct solution for the whole cube is obtained, which corresponds to Fig.5(d). Note that there is no corresponding zero-cost solution for Fig.5(b).